MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

Bachelor of Science (Honours)Biotechnology

COURSE CODE – MG1DSCBTG100

Course Title - Fundamentals for Biotechnology

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer the questions in one word or sentence
Answer All Questions
Each question carries 1 mark

1. Write the meaning of the word "**cyto**" [K] [CO1].
2. Who proposed the cell doctrine? [K] [CO1].
3. Name the type of ribosome is present in eukaryotic cell. [K] [CO1].
4. The process of transferring a nucleus to a new cell is called---- [U] [CO1].
5. The basic unit of DNA is called ------ [U] [CO1].
6. The Genetic makeup of an organism is? [U] [CO1].
7. Common cold is caused by what type of microbes? [K] [CO1].
8. The condition when body attacks it's own tissues and cells is known as? [U] [CO1].
9. DNA sequencing was first done by---- [K] CO1]
10. Which immunoglobulin can cross through placenta? [K] [CO1]. **(1 X 10 = 10 Marks)**

Short answer questions
Answer any 4 questions.
Each question carries 3 marks.

11. Explain the significance of Schleiden and Schwann's contributions to cell theory. [U] [CO 2)].
12. What is gene therapy? [U] [CO2].
13. What is the role of *Agrobacterium tumefaciens* in plant biotechnology? [U] [CO2].
14. Explain the denaturation step in PCR. [U] [CO2].
15. Analyse the law of independent assortment with segregation of gametes. [An] [CO3].
16. Solve the issue of entry of pathogen to animal cells which causes a disease. How the cell fights against. [Apply] [CO 3]. **(3 X 4 = 12 Marks)**

Short Essay questions
Answer any 3 questions.
Each question carries 6 marks.

17. Describe the monohybrid cross carried out by Mendel. [K] [CO1].
18. Differentiate the structure of DNA and RNA. [An] [CO3].
19. Outline the milestones in the development of Biotechnology. [U] [CO2].
20. rDNA technology is used to produce Genetically modified organisms. Justify. [An] [CO3].
21. Describe the general characteristics of Virus. [U] [CO2]. **(6 X 3 = 18 Marks)**

Essay questions
Answer any 1 question. (10 marks)

22. Give an outline of rDNA technology. [U] [CO2].
23. Immunological responses are necessary for get rid of diseases. Outline the mechanism in detail. [An] [CO3]. **(10 X 1 = 10 Marks)**